B. TECH-ME-III-II Sem

23A03605c	DESIGN OF AUTOMOBILE TRANSMISSION SYSTEMS **(Professional Elective-III)**	L	T	P	C
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Course objectives: The objectives of the course are to	
1	Explain the various elements involved in a transmission system.
2	Focus on the various forces acting on the elements of a transmission system.
3	Design the system based on the input and the output parameters.
4	Produce working drawings of the system involving pulleys, gears, clutches and brakes.
5	Demonstrate the energy considerations in the design of motion control elements.

COURSE OUTCOMES On successful completion of this course the student will be able to		
1	Design and select suitable flexible power transmission elements and analyze the load conditions, and performance requirements.	*L2,L4,L6*
2	Analyze spur gear geometry and kinematics, and design spur gears for desired power transmission capacity based on bending and contact stress criteria.	L1,L4,L6
3	Analyze and design different types of gears by evaluating their efficiency, ensuring reliable performance in mechanical power transmission systems.	L4,L5,L6
4	Create speed reducers and multi-speed gearboxes by developing structural and ray diagrams, selecting appropriate gear combinations for various mechanical applications.	L2,L3,L6
5	Select, and design various types of clutches and brakes, analyze and choose appropriate friction materials for effective motion control.	L2,L4,L6

UNIT I

Flexible power transmission systems: Design of Belts – Flat Belts and Pulleys – V Belts and Pulleys – Design of chain drives – Wire ropes

Design of bearing: Lubrication- hydrodynamic lubrication theory, Design of sliding contact bearing using Sommer field number – Design using Mckee's equation – Selection of rolling contact bearings.

UNIT II

Spur gear: Gear geometry – Kinematics – Forces on gear tooth – Stresses in Gear tooth – Selection of gear material based on bending stress and contact stress – Design of Spur gear – Power transmitting capacity.

UNIT III

Helical, bevel and worm gears: Parallel Helical Gears – Kinematics – Tooth proportions – Force analysis – Stresses in Helical gear – Design of helical gear – Crossed Helical gears – Straight Bevel gears – Kinematics – Force analysis – Stresses in straight bevel gear tooth – Design of bevel gear – Worm gearing – Kinematics – Forces - Friction and Efficiencies – Stresses in worm gear tooth.

UNIT IV

Design of gear boxes: Design of Speed reducers – Design of multi speed gear boxes for machine tools – Structural and ray diagrams.

UNIT V

Elements of motion control: Internal – Expanding Rim clutches and Brakes – External – Contracting Rim clutches and Brakes – Band type Clutches – Cone clutches and Brakes – Energy considerations – Temperature rise – Friction materials.

TEXT BOOKS:

1. Joseph Edward Shigley and Charles, R. Mischke, "Mechanical Engineering Design", McGraw –Hill International Editions, 2000.
2. Robert L. Norton, "Machine Design"- an integrated approach, (5th Edition) Pearson publisher, 2000

REFERENCES:

1. "Design Data", PSG College of Technology, DPV Printers, Coimbatore, 2005.
2. Malisa, "Hand Book of Gear Design", Tata Mc Graw Hill, International Edition, 2000.
3. V.B. Bhandari ,"Design of Machine Elements", Tata Mc Graw Hill, 2001

Online Learning Resources:

https://youtube.com/playlist?list=PLyqSpQzTE6M-7nTyaGekZRTLLUzGfRPMo&si=Jvicxjkhv8LS6Lt_

https://youtube.com/playlist?list=PLyqSpQzTE6M-7nTyaGekZRTLLUzGfRPMo&si=aFp27b3qPylydjCV

https://youtu.be/ftJKqKuppF4?si=wzfkYJUOeDxWHWRW